

**PROPOSAL FOR THE IMPACT OF IMMERSIVE TECHNOLOGIES
ON STEM EDUCATION IN SRI LANKA'S SECONDARY AND
TERTIARY CURRICULUM**

25-26J-418

Project Proposal Report

D.A.M.I.G. Andradi

IT22923592

B.Sc. (Hons) Degree in Information Technology Specializing in
Interactive Media

Department of Information Technology
Sri Lanka Institute of Information Technology, Sri Lanka

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Signature
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Signature of the Supervisor:

Date:

Signature of the Co-Supervisor:

Date:

ABSTRACT

In the context of Sri Lankan STEM education, secondary schools often teach ICT concepts, such as data flow, network topologies, and logic gates, using static textbook diagrams. This approach emphasizes rote memorization rather than conceptual understanding, particularly in schools lacking adequate laboratory facilities. To address these limitations, this study presents the development of a marker-based Augmented Reality (AR) mobile application that transforms Grade 10 ICT textbook diagrams into interactive 3D models accessible via smartphones. Unlike prior AR educational interventions, which have largely focused on foreign contexts or high-end hardware, this Unity-based application is optimized for low-specification Android devices and functions offline with existing government-issued Sinhala-language textbooks. A quasi-experimental design across multiple Sri Lankan schools will be used to evaluate the app's impact on students' conceptual understanding of ICT topics, student engagement, and teacher adoption of the AR application. The study aims to provide practical AR implementation strategies for resource-constrained educational environments and to contribute empirical evidence regarding the effectiveness of AR in STEM education in developing countries, particularly within the context of Sri Lankan secondary education.

Keywords: Augmented Reality, STEM Education, Mobile Application, Computer Vision, IT Curriculum.

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List of Abbreviations

Abbreviations	Descriptions
AR	Augmented Reality
VR	Virtual Reality
MR	Mixed Reality
ICT	Information and Communication Technology
IT	Information Technology
R&D	Research and Development
STEM	Science, Technology, Engineering, and Mathematics
UI/UX	ser Interface and User Experience
APIs	Application Programming Interfaces

1 INTRODUCTION

The teaching of Information and Communication Technology (ICT) in secondary schools presents several pedagogical challenges. Core concepts such as data communication, network design, binary logic, and hardware interaction require learners to construct mental models of systems that are inherently abstract and often invisible. Traditional textbook diagrams, however, remain static and limited in their ability to depict the dynamic flow of information. Within the Sri Lankan context, where ICT is a compulsory subject at Grade 10, but computer laboratory facilities are often insufficient, students face difficulty in connecting theoretical knowledge to practical application. This situation frequently results in surface-level learning rather than the development of deep conceptual understanding.

Over the past two decades, immersive technologies such as Augmented Reality (AR) and Virtual Reality (VR) have emerged as promising approaches to addressing these pedagogical limitations. Batista et al. [2] conducted a comprehensive systematic review of 14 studies published between 2010 and 2018 that employed AR, VR, and Mixed Reality (MR) in computer science education. Their findings indicated consistently positive outcomes, including increased student motivation (reported in six studies), improved learning efficiency (nine studies), enhanced understanding (four studies), and greater interactivity (three studies). The review further revealed that discovery-based learning strategies dominated the field (79% of studies), underscoring the value of AR/VR technologies in facilitating active exploration and knowledge construction. Applications covered a broad spectrum of subject matter, ranging from programming logic and algorithms to network security and data structures, demonstrating the versatility of these technologies for diverse computer science domains.

Building on this foundation, Taufik et al. [4] designed a marker-based AR mobile application aimed at supporting computer network hardware education in Indonesian vocational schools. Using a Research and Development (R&D) methodology, their study involved 72 students across experimental and control groups. The intervention demonstrated statistically significant improvements, with the experimental group achieving an average post-test score of 88.44 compared to 80.69 in the control group ($p < 0.05$). Moreover, usability testing conducted in accordance with ISO 25010 standards yielded high satisfaction ratings from media experts (82.4%), subject matter experts (83.75%), teachers (83.85%), and students (83.82%). The normalized gain (N-Gain) score of 77.1078 exceeded the threshold for effectiveness, confirming that AR technology can enhance the visualization of complex networking hardware concepts and support improved learning outcomes.

The technical feasibility of AR applications for network visualization was further demonstrated by Sato et al. [1], who developed a platform for managing wireless sensor networks through AR technology. Their system allowed users to overlay complex network topologies onto real-world environments using smartphone cameras, while simultaneously abstracting multiple

communication protocols (Wi-Fi, Bluetooth, ZigBee) through unified APIs. Performance testing revealed response times of 0.1–0.5 seconds for information retrieval and 2–3 seconds for device control, confirming the potential of AR for real-time applications. Importantly, their comparative evaluation highlighted the superiority of AR-based visualization over conventional 2D or 3D approaches, as AR more effectively bridged the gap between virtual representations and physical device placements.

Cowling and Birt [3] provided valuable pedagogical insights through a mixed reality intervention combining 3D-printed physical network components with mobile AR simulations. Their pilot study demonstrated high usability, with students rating memorability (5.00) and navigability (4.33) particularly favorably. However, the study also revealed key instructional challenges. Students tended to bypass the intended two-stage process—physical construction followed by AR visualization—opting instead to experiment directly with AR. This unstructured exploration created difficulties in comprehending more complex networks, underscoring the importance of scaffolding and guided learning pathways in AR/MR applications.

Despite the demonstrated benefits of these studies, several limitations remain evident. Many interventions target advanced or vocational contexts, such as Taufik et al.'s focus on vocational school students or Cowling and Birt's work with undergraduates, rather than secondary-level learners. Furthermore, several applications assume the availability of mid- to high-end devices or specialized hardware, conditions that do not reflect the realities of resource-constrained Sri Lankan classrooms. Sato et al.'s platform, for example, required compatibility with multiple wireless standards and advanced smartphone capabilities. In addition, while Batista et al. [2] identified discovery-based strategies as predominant, none of the reviewed studies integrated AR directly into government-issued textbooks or assessed adoption by classroom teachers.

In response to these gaps, the present proposal seeks to develop a marker-based AR application that integrates directly with Grade 10 ICT textbooks in Sri Lanka. Drawing inspiration from the marker-based methodology of Taufik et al. [4] and incorporating the scaffolded instructional design principles identified by Cowling and Birt [3], the application will enable students to scan textbook diagrams and view corresponding 3D simulations, such as data flow within a network topology or signal transmission through logic gates. The evaluation will employ the mixed-methods approach recommended by Batista et al. [2], combining quantitative measures of learning outcomes with qualitative assessments of user satisfaction and teacher adoption. In doing so, this research aims to contribute both practical strategies for improving ICT education in resource-constrained environments and empirical evidence to the broader literature on the integration of AR in STEM education.

1.1 Background

The integration of immersive technologies in computer science education has gained increasing attention over the past decade, particularly as a response to the pedagogical challenges associated with teaching abstract computational concepts. Traditional methods often rely on static representations, which limit students' ability to visualize and interact with complex processes. Augmented Reality (AR), Virtual Reality (VR), and Mixed Reality (MR) have emerged as promising tools to address these challenges, offering interactive, exploratory, and experiential learning opportunities. While AR and VR applications have been shown to enhance engagement, motivation, and conceptual understanding in various educational contexts, their adoption within secondary-level ICT education remains limited, particularly in resource-constrained environments such as Sri Lanka. Furthermore, despite significant global advancements, there remains a lack of context-specific research focusing on the unique infrastructural and pedagogical challenges present in South Asian educational systems.

1.2 Literature Survey

A growing body of research highlights the potential of immersive technologies in computer science and ICT education. Batista et al. [2] conducted a systematic review of 14 studies published between 2010 and 2018 that applied AR, VR, and MR in computer science education. Their analysis revealed positive learning outcomes across multiple dimensions, with VR being the most widely employed technology (64.29%), followed by AR (35.71%). Notably, no studies implemented MR applications. Unity3D was identified as the predominant development platform, while AR-specific tools such as ARToolKit and Aurasma were also employed. The reviewed applications addressed both foundational topics (e.g., programming logic and algorithms) and advanced subjects (e.g., artificial intelligence, network security, and data structures). Instructionally, most applications (79%) adopted discovery-based learning approaches, while evaluation methodologies varied, including qualitative (29%), quantitative (22%), and mixed methods (21%). Importantly, the review highlighted significant gaps, including the underrepresentation of South Asian contexts and the absence of MR-based applications.

Expanding upon this, Taufik et al. [4] examined the practical use of AR for computer network hardware education in Indonesian vocational schools. Their Android-based, marker-driven AR application enabled students to visualize and interact with 3D models of network hardware that were otherwise inaccessible in resource-limited environments. The application was developed using the Borg and Gall R&D model and evaluated through ISO 25010 standards, achieving high satisfaction ratings from media experts (82.4%), subject experts (83.75%), teachers (83.85%), and students (83.82%). Effectiveness testing, employing Independent Sample T-tests and N-Gain calculations, demonstrated statistically significant improvements in learning outcomes, with the experimental group achieving higher post-test scores (88.44) than the control group (80.69). However, the study was limited to a single subject domain and an Indonesian vocational education context, leaving questions regarding broader curricular applicability.

Further exploration of AR's technical feasibility was presented by Sato et al. [1], who developed an AR-based platform for wireless sensor network visualization and management. The system supported multiple communication standards (Wi-Fi, Bluetooth, ZigBee) through unified APIs and employed smartphone cameras to overlay network topology information in real-world environments. Evaluation results demonstrated low latency (0.1–0.5 seconds for retrieval, 2–3 seconds for device control), confirming the feasibility of real-time AR applications. The platform offered key visualization features, including link visualization, traffic visualization, and device information, alongside network control capabilities. Comparative analysis with existing visualization tools indicated that AR was more effective in bridging the gap between virtual representations and physical device placement.

Pedagogical insights into AR/MR applications were provided by Cowling and Birt [3], who piloted an intervention combining 3D-printed physical network components with mobile AR simulations. Their mixed reality approach fostered a two-stage learning process, enabling students to first construct physical networks and then visualize packet flow using AR. Usability testing with undergraduate students yielded high scores for accessibility (4.67), learnability (4.00), memorability (5.00), and navigability (4.33). Nonetheless, the study identified critical instructional design challenges. Students frequently bypassed the intended two-stage process, engaging prematurely with AR simulations, which created difficulties in comprehending complex networks. The findings underscored the necessity of structured scaffolding and guided learning pathways in AR/MR-based education.

Taken together, these studies collectively highlight the pedagogical and technical potential of immersive technologies in ICT education. However, they also reveal notable gaps: the absence of comprehensive MR applications in secondary education, the concentration of studies in vocational or higher education contexts, and the lack of integration with standard school curricula and textbooks. Most importantly, there remains a significant research void in South Asian settings, including Sri Lanka, where infrastructural constraints, limited device availability, and curriculum-specific requirements pose unique challenges. Addressing this gap represents a critical opportunity to adapt and evaluate immersive technologies in resource-constrained educational environments, thereby extending their proven benefits to secondary-level ICT education in the region.

1.3 Research Gap

Although prior research demonstrates the promise of Augmented Reality in ICT education, several critical gaps remain unaddressed when examined through the lens of practical implementation in developing educational contexts. A comprehensive analysis of existing literature reveals four primary areas where current research falls short of addressing real-world educational needs: curricular alignment gaps, technological accessibility gaps, geographical and contextual gaps, and pedagogical integration gaps.

Curricular Alignment Gap

The most significant limitation in existing research is the mismatch between developed applications and actual secondary school curricula. Taufik et al. [4] developed their AR application specifically for vocational school students in the Computer Engineering and Networking (TKJ) department, focusing on advanced network hardware concepts such as routers, switches, access points, and network topology design. Their curriculum targeted specialized technical education with students already committed to ICT careers, involving complex networking concepts that extend far beyond typical secondary school ICT education. The study's content scope included detailed examination of network devices, cable types, IP addressing schemes, and network configuration procedures—topics that are either absent from or only superficially covered in general Grade 10 ICT curricula.

Similarly, Cowling and Birt [3] designed their mixed reality intervention for undergraduate university students studying computer networking, with content focused on advanced TCP/IP model visualization, packet routing algorithms, and multi-layer network protocol analysis. Their application assumed prior knowledge of networking fundamentals and was designed to enhance understanding of complex theoretical multi-step problems that are typically encountered in tertiary-level computer science programs. The study's target audience possessed technical backgrounds and learning objectives that align with professional ICT training rather than foundational secondary education.

Batista et al.'s systematic review [2] further highlighted this curricular misalignment by revealing that most AR/VR applications in computer science education targeted advanced topics such as artificial intelligence, network security, data structures, and algorithm optimization. Among the 14 reviewed studies, content coverage ranged from programming logic and computational thinking to specialized areas like search algorithms and binary search techniques. Notably, none of the applications were designed to align with standard secondary school ICT curricula that typically emphasize digital literacy, basic computer operations, introductory programming concepts, and fundamental understanding of information systems rather than advanced technical implementations.

Technological Accessibility Gap

Current research consistently assumes access to high-specification hardware and robust technological infrastructure that are not representative of resource-constrained educational environments. Sato et al. [1] developed their AR platform using high-end development environments including Intel Core i7 processors with 8GB RAM, professional-grade wireless communication modules (SunSPOT base stations), and specialized development frameworks requiring substantial computational resources. Their implementation utilized multiple wireless communication standards simultaneously (Wi-Fi, Bluetooth, ZigBee) and required real-time processing capabilities that demand considerable system resources and stable network connectivity.

Taufik et al.'s study [4] did not explicitly specify minimum hardware requirements for their Android-based AR application, but their implementation details suggest assumptions about device capabilities that may not be universally available. Their use of marker-based recognition, 3D models rendering, and real-time overlay processing typically requires devices with sufficient processing power, camera resolution, and memory allocation that exceed the specifications of entry-level smartphones commonly available in developing countries. The study's evaluation was conducted in a controlled school environment with presumably adequate infrastructure, but no consideration was given to varying device capabilities or network connectivity limitations.

Cowling and Birt [3] utilized Unity3D development platform with Vuforia AR plug-in, technologies that require substantial development resources and result in applications with significant memory and processing requirements. Their mixed reality approach combining physical 3D-printed components with digital AR simulations assumed access to both specialized hardware (3D printers for physical models) and capable mobile devices for AR visualization. The study's implementation environment suggested institutional support and resources that may not be available in typical secondary school settings.

The systematic review by Batista et al. [2] revealed that 35.71% of reviewed applications used Unity3D as their primary development platform, with additional reliance on specialized AR frameworks like ARToolKit and Vuforia. These development choices typically result in applications with substantial resource requirements that may not be compatible with lower-specification devices. The review noted that most studies did not address device compatibility or performance optimization for resource-constrained environments, representing a significant gap in practical applicability.

Geographical and Contextual Gap -Perhaps the most critical gap identified in current research is the absence of studies conducted within South Asian educational contexts, particularly Sri Lankan schools. Batista et al.'s comprehensive systematic review [2] analyzed 14 studies published between 2010-2018 and found minimal representation from South Asian countries, with research predominantly originating from developed countries with substantially different educational infrastructure, technological access, and pedagogical approaches. The reviewed studies were conducted in contexts where schools typically have access to computer laboratories, high-speed internet connectivity, and institutional support for technology integration—conditions that do not reflect the reality of many South Asian educational environments.

Taufik et al.'s research [4], while conducted in Indonesia and therefore geographically relevant to the Southeast Asian region, was implemented in SMK Negeri 3 Makassar, a state vocational school with presumably adequate technological infrastructure and institutional support for educational innovation. The study did not address challenges commonly faced in resource-constrained environments such as inconsistent electricity supply, limited internet connectivity, varied device availability among students, or institutional resistance to technology adoption. Furthermore, the vocational school context provided specialized ICT facilities and technically oriented students that differ significantly from general secondary school environments.

Sato et al. [1] and Cowling and Birt [3] conducted their research in developed country contexts (Japan and Australia respectively) with access to well-equipped university laboratories, stable technological infrastructure, and institutional support for experimental educational technologies. These studies did not consider implementation challenges related to limited resources, infrastructure constraints, or the need to integrate with existing educational materials and practices commonly found in developing educational systems.

Pedagogical Integration Gap -A critical oversight in existing research is the failure to address integration with existing educational materials and practices. None of the reviewed studies designed their AR applications to work directly with government-issued textbooks or existing curricular resources. Taufik et al. [4] created custom marker-based materials specifically for their AR application, requiring schools to adopt new educational resources rather than enhancing existing ones. Their approach necessitated printing specialized markers and developing entirely new instructional materials, creating additional costs and implementation barriers for schools with limited budgets.

Cowling and Birt [3] required 3D-printed physical components that schools must manufacture or purchase separately from their AR application. Their mixed reality approach, while pedagogically sound, created dependency on specialized hardware that schools must acquire and maintain in addition to their existing educational resources. The study did not evaluate how their intervention could be integrated with existing networking education materials or adapted to work within current pedagogical frameworks.

Sato et al.'s platform [1] was designed as a professional network management tool rather than an educational application, requiring technical expertise for implementation and maintenance that extends beyond typical teacher capabilities. Their system assumed institutional technical support and specialized training that may not be available in standard secondary school environments.

Furthermore, none of these studies addressed teacher training requirements, institutional adoption challenges, or long-term sustainability considerations that are crucial for successful educational technology implementation. The gap between proof-of-concept research and practical classroom implementation represents a significant barrier to realizing the demonstrated benefits of AR technology in ICT education, particularly in resource-constrained environments where institutional support for technology integration may be limited.

This comprehensive analysis of existing research gaps provides strong justification for developing an AR application specifically designed for Sri Lankan Grade 10 ICT education that addresses curricular alignment, technological accessibility, contextual appropriateness, and pedagogical integration requirements.

1.4 Research Problem

The central problem this study seeks to address is how an Augmented Reality (AR) system can support Sri Lankan secondary school students in moving from passive consumption of static textbook diagrams to active engagement with complex STEM concepts. The aim is to design a technological platform that transforms abstract Information and Communication Technology (ICT) content into tangible and interactive experiences, thereby fostering deeper conceptual

understanding. This problem stems from a widely documented pedagogical challenge: the persistent disconnect between theoretical knowledge presented in static formats and the dynamic, interconnected processes that characterize ICT systems.

Batista et al. [2] identified this issue in their systematic review, which examined the application of AR and VR in computer science education. Their findings indicated that 79% of effective applications employed discovery-based learning strategies, allowing students to actively construct knowledge through exploration rather than through passive information absorption. The review further demonstrated that traditional teaching approaches often create a “mental gap” when students attempt to understand concepts without direct physical analogues—an especially critical issue in ICT education, where phenomena such as data flows, network protocols, and system architectures are inherently invisible. Immersive technologies were shown to alleviate this challenge by improving learning facilities (reported in nine of the reviewed studies) and enhancing conceptual understanding (four studies), primarily by enabling students to manipulate and explore interactive mental models.

The urgency of this pedagogical challenge is particularly evident in the Sri Lankan context. While Taufik et al. [4] provide a relevant methodological precedent through their Indonesian study, their focus on vocational education highlights a gap in application. Their AR intervention produced statistically significant improvements in student performance (experimental group mean score of 88.44 versus control group mean score of 80.69), demonstrating that abstract networking concepts could be effectively taught through interactive visualizations. However, their work was limited to specialized vocational contexts rather than general secondary ICT curricula. The success of their application, which enabled students to interact with realistic 3D representations of networking components, provides strong evidence that transitioning from 2D to 3D interactive representations directly addresses comprehension barriers. Nevertheless, there remains a need to adapt this approach to foundational ICT education in resource-constrained secondary school environments.

The technical feasibility of real-time AR visualization of abstract processes was further validated by Sato et al. [1]. Their platform, designed for wireless sensor networks, enabled users to visualize network topologies, data transmission, and device interactions through AR overlays. With response times of 0.1–0.5 seconds for information retrieval and 2–3 seconds for device control, their system demonstrated that AR can effectively support interactive, real-time engagement with complex processes. Such evidence confirms the technological viability of AR in creating responsive educational environments that allow immediate feedback and reinforce conceptual understanding.

However, Cowling and Birt [3] highlighted a crucial pedagogical consideration in their mixed reality intervention for networking education. While their study achieved high usability scores (memorability 5.00; navigability 4.33), it also revealed that unguided exploration often resulted in confusion and limited learning effectiveness. This finding emphasizes that technological capacity alone is insufficient; AR applications must incorporate structured instructional design with scaffolding that guides learners progressively from surface-level familiarity toward deeper conceptual mastery.

The research problem therefore encompasses both technological and pedagogical dimensions. On the one hand, there is a need to develop AR systems capable of bridging the representational gap between static 2D diagrams and dynamic 3D conceptual models. On the other hand, these systems must be embedded within pedagogically sound frameworks that provide structured learning pathways and ensure meaningful progression of understanding. Within Sri Lanka, these challenges are further compounded by infrastructural limitations, device availability, and the necessity of aligning with government-issued textbooks. Consequently, the research problem lies in creating AR solutions that are technologically sophisticated, pedagogically grounded, and practically implementable within the constraints of the Sri Lankan secondary education system.

2 OBJECTIVES

2.1 Main Objectives

The primary objective of this research is to design, develop, and empirically evaluate a marker-based Augmented Reality (AR) mobile application aimed at enhancing STEM education for Grade 10 students in Sri Lankan secondary schools. Specifically, the application seeks to transform static textbook diagrams into interactive three-dimensional learning experiences, thereby addressing the pedagogical challenge of making abstract Information Technology (IT) concepts more accessible and comprehensible through immersive visualization. By doing so, the study intends to bridge the gap between theoretical knowledge and practical understanding in ICT education.

The proposed AR application will be engineered to integrate seamlessly with government-issued Sinhala-language textbooks. Using image recognition, the system will detect diagrams related to network topologies, data flow processes, logical circuits, and system architectures. Once scanned, the application will generate interactive 3D models that dynamically represent processes such as packet transmission across networks, signal propagation within digital circuits, and data transformation in computing systems. This approach complements rather than replaces existing educational resources, ensuring feasibility within Sri Lanka's current financial and infrastructural constraints.

A mixed-methods evaluation framework will be employed to assess the effectiveness of the intervention. Quantitative measures, including pre- and post-intervention assessments, will capture improvements in student comprehension of ICT concepts, while qualitative measures, such as usability testing and focus group feedback, will evaluate student engagement, motivation, and autonomous learning potential. In addition, teacher feedback will be systematically gathered to identify challenges and opportunities for pedagogical integration.

Performance optimization will be prioritized to ensure functionality on mid-range Android devices (2–3 GB RAM), thereby supporting accessibility across diverse socioeconomic contexts. The application will be directly aligned with the Grade 10 ICT curriculum, focusing on challenging topics such as computer generations, network topologies, logic gates, and input/output devices. By targeting concepts that are traditionally difficult to grasp through static textbook representations, the study aims to deliver measurable improvements in conceptual understanding while providing a scalable, resource-sensitive solution for STEM education in Sri Lankan classrooms.

2.2 Specific Objectives

The specific goals for the Information Technology Lead part of the project are a set of clear goals that are essential for the entire project's success. The IT Lead is responsible for the main technological part of the app, which provides the base for all the educational content created by the other team members.

1. **To design and create a strong mobile app platform** that can recognize and track specific diagrams in government-issued Sinhala textbooks in real time. This goal can be measured by how fast and accurate the image recognition system is, and it is focused on the project's specific audience and materials.
2. **To develop a fast 3D graphics system** for mobile devices to make sure the Mixed Reality experience is smooth and doesn't lag. This is important because a slow app can make students frustrated and less interested in learning.
3. **To create a simple and easy-to-use user interface and experience (UI/UX)** that allows students to naturally interact with 3D models and simulations. The app's design must be simple enough for a Grade 10 student to use without needing a lot of instructions.
4. **To design and create interactive visuals for specific abstract IT concepts** from the Grade 10 curriculum, like how data packets move in a network and how logical gates work. This goal solves the unique challenge of making invisible processes visible, which is a main part of this project's new approach.

3 METHODOLOGY

3.1 Overall System Description

This study adopts a marker-based Augmented Reality (AR) approach, building on prior evidence that demonstrates the effectiveness of such systems in transforming static educational resources into interactive learning experiences. Taufik et al. [4] successfully employed marker recognition to trigger 3D visualizations of networking components, reporting significant improvements in learning outcomes and high student satisfaction (83.82%). Extending this approach, the proposed system will utilize existing Grade 10 ICT textbook diagrams as natural markers, eliminating the need for additional printed materials and thereby reducing implementation costs for schools.

System Architecture

The technical architecture is informed by Sato et al. [1], whose AR platform achieved real-time visualization of network processes through mobile devices with response times of 0.1–0.5 seconds. Unlike their professional-grade implementation requiring high-end hardware, the proposed system will be optimized for mid-range Android devices with 2–3 GB RAM. This ensures accessibility for Sri Lankan students while maintaining interactive responsiveness.

Marker recognition will function by matching pre-registered textbook diagrams in the application database. When a diagram is scanned, computer vision algorithms will identify its orientation, enabling the accurate spatial overlay of corresponding 3D models. For example, a static diagram of a network topology will be enhanced with animated packet transmission between nodes, with interactive features allowing students to explore device roles and functions.

Pedagogical Design

Drawing on the findings of Cowling and Birt [3], who highlighted the risks of unguided exploration in mixed reality learning, the system incorporates structured scaffolding mechanisms. Visualizations will employ progressive disclosure: initially presenting simplified representations of ICT concepts, followed by layered details as student understanding deepens. This ensures that AR content supports learning progression rather than overwhelming learners.

- Interactive modules will be tailored to specific ICT topics:
- Network topologies – animated simulations of data flow.
- Computer hardware – 3D models with removable components.
- Logic circuits and processes – step-by-step execution sequences.

Each visualization will be supported by contextual information presented in Sinhala to ensure accessibility for students with varying English proficiency.

Performance Optimization

Technical considerations identified in Batista et al.'s [2] systematic review inform the optimization strategies of the proposed system. To balance visual complexity with hardware limitations, the Unity3D platform will be used with adaptive rendering techniques that automatically adjust visual quality based on device performance. This ensures smooth operation on low-specification devices common in Sri Lankan classrooms.

Environmental challenges such as lighting variability will be addressed through robust computer vision algorithms capable of recognizing markers under diverse illumination conditions. A calibration feature will further enable users to optimize recognition sensitivity. Additionally, offline functionality will ensure continuous access to AR resources in rural or infrastructure-limited contexts.

Integration with Educational Resources

A key innovation of this research lies in its integration with government-issued textbooks. While previous studies required custom-printed markers [4] or physical components [3], the proposed system leverages diagrams already embedded in the standard curriculum. This design choice minimizes additional costs and maximizes scalability by aligning with existing educational infrastructure, ensuring both feasibility and sustainability of implementation within Sri Lankan secondary schools.

3.2 System Description

1) Input Layer (What the App Sees)

- Students point their phone camera at specific pictures or diagrams in their textbooks
- These textbook images act like "triggers" that the app recognizes (called AR markers)
- The app needs clear, high-quality images to work properly
- Each image is linked to a specific 3D model or animation in the app's database

2) Processing Layer (What Happens Behind the Scenes)

- The app uses Unity game engine (a popular tool for creating interactive apps) combined with AR Foundation (Apple and Google's AR technology)
- When the camera recognizes a textbook image, the app quickly loads the matching 3D content
- All 3D models are simplified (low poly) so they run smoothly on regular smartphones without draining the battery

- The app calculates where to place the 3D objects, so they appear to sit perfectly on top of the textbook page
- 3) Output Layer (What Students See and Do)
- Interactive 3D models appear on the phone screen, overlaid on the real textbook
 - Examples: A flat circuit diagram becomes an animated simulation showing electricity flowing, or a network diagram shows data packets moving between computers
 - Students can rotate, zoom, or interact with these 3D objects using touch gestures
 - The virtual objects stay locked to the textbook page even when students move their phone around
- 4) Data Logging (How We Measure Success)
- The app secretly records how students use it (with permission)
 - Tracks which feature students use most, how long they spend with each concept, and what problems they encounter
 - This data helps improve the app and proves whether it helps students learn better.

3.3 Technologies to be Used

Unity 3D Game Engine Foundation the Augmented Reality application will be built using Unity 3D, a widely used game development platform that supports both Android and iOS devices. Unity was chosen because it provides cross-platform compatibility, meaning one codebase can work on different mobile operating systems without requiring separate development for each platform. Unity's XR Toolkit offers pre-built components for augmented reality development, including camera access, object tracking, and real-time 3D rendering capabilities. This foundation allows the development team to focus on educational content creation and learning experience design rather than building core AR functionality from scratch.

Image Recognition and Tracking System A robust computer vision system forms the technical backbone of the application, responsible for recognizing textbook diagrams and maintaining stable tracking as students move their devices. The system must identify specific images from government-issued textbooks under varying conditions including different lighting environments, partial occlusions when students' hands cover parts of pages, and various viewing angles. Technology options under consideration include Vuforia (known for reliable marker tracking), ARCore (Google's AR platform), and ARKit (Apple's AR framework). The final selection will depend on tracking accuracy, mobile processor efficiency, and performance consistency across the range of smartphones commonly used by Sri Lankan students.

Mobile Graphics Optimization Ensuring smooth performance on mobile devices is critical for maintaining student engagement and learning effectiveness. Poor performance, including lag, stuttering, or crashes, can break the immersive learning experience and lead to student frustration. The graphics optimization strategy includes implementing efficient shaders (programs that control how 3D objects appear), compressed textures to reduce memory usage, and level-of-detail systems that automatically simplify 3D models when viewed from distance.

Additionally, the system will use occlusion culling to avoid rendering objects not visible to the camera, and batch rendering techniques to reduce processing overhead. These optimizations ensure the application runs smoothly on budget smartphones typically available to students, maintaining consistent frame rates and responsive interactions.

Data Collection and Analytics Framework An integrated analytics system will monitor student interactions, tracking metrics such as session duration, feature usage frequency, learning module completion rates, and user interface navigation patterns. This data collection serves dual purposes: providing insights for iterative application improvement and generating quantitative evidence of educational effectiveness for research validation.

3.4 System Architecture Diagram

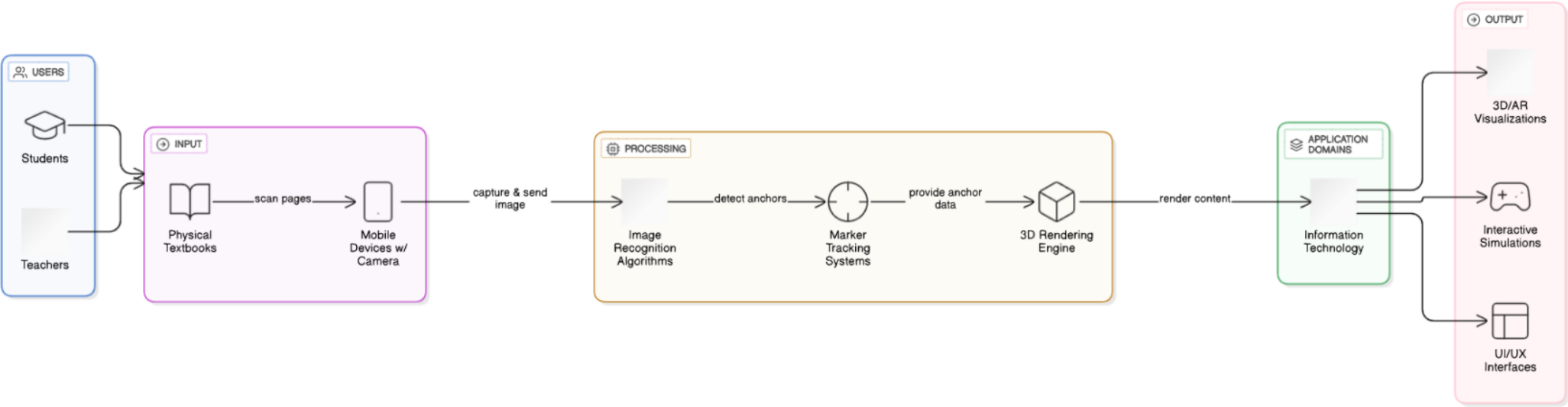


Figure 1: System Architecture Diagram

3.5 High level Architecture Diagram

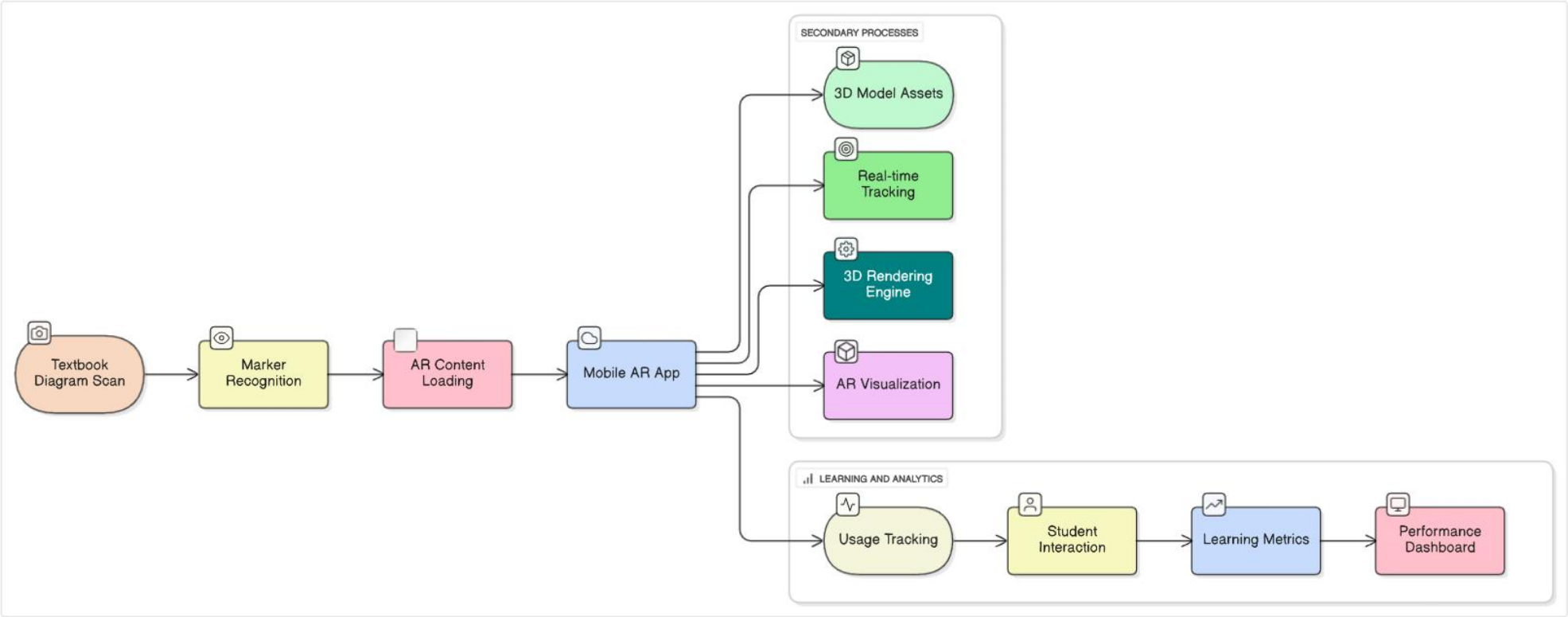


Figure 2: High level Architecture Diagram

3.6 Gantt Chart

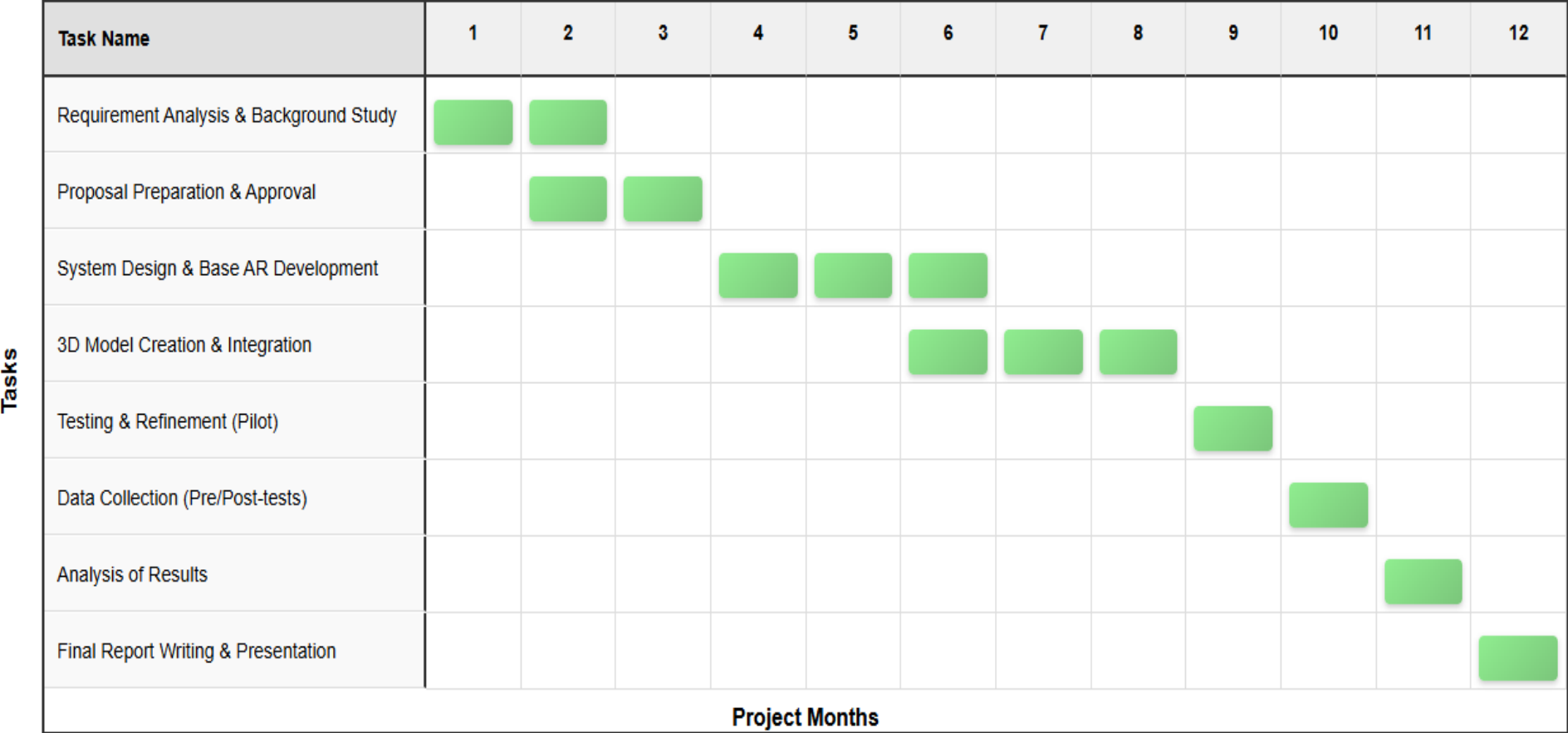


Figure 3: Gantt Chart

3.7 Work Breakdown Structure

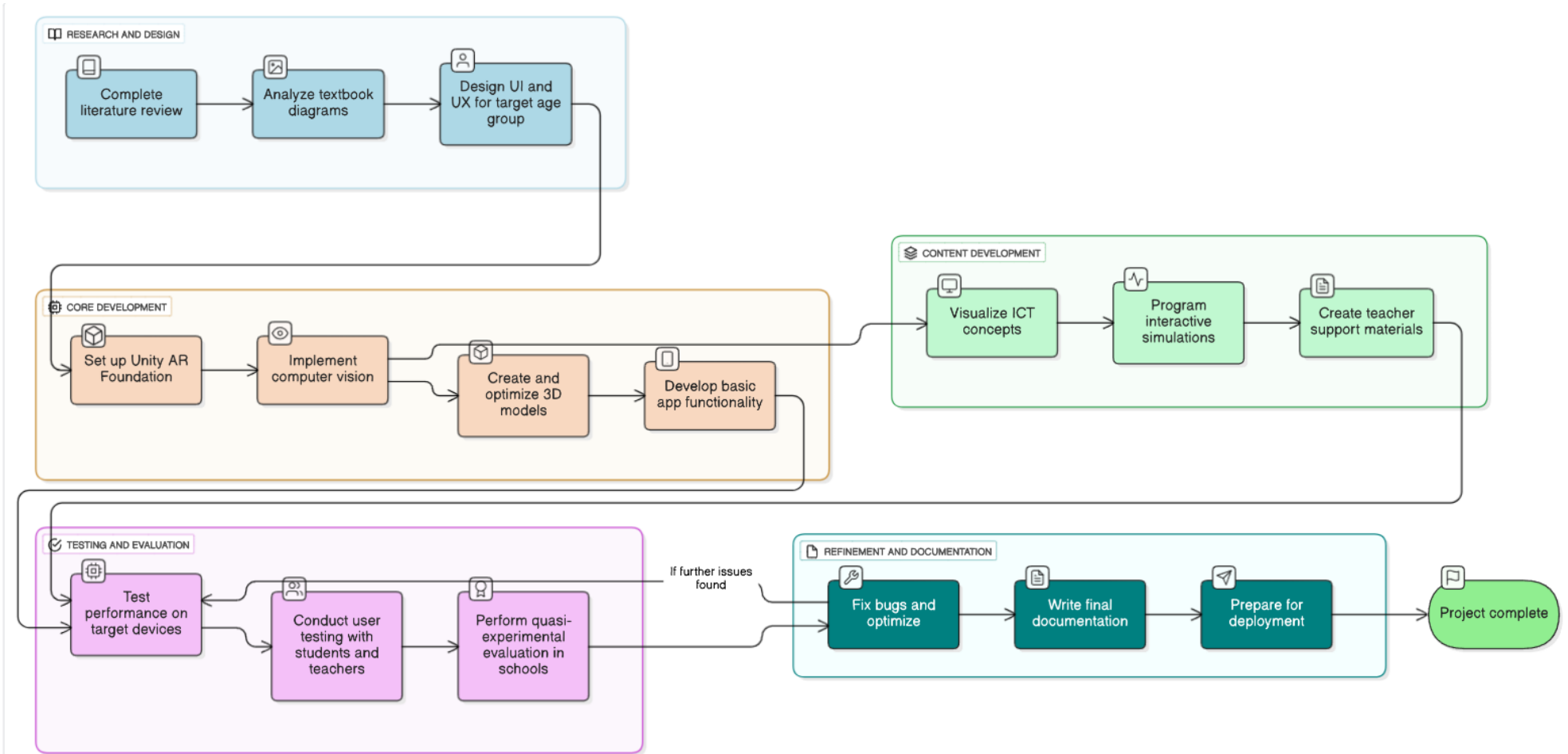


Figure 4: Work Breakdown Chart

4 PROJECT REQUIREMENTS

4.1 Functional Requirements

Functional requirements define what the system must do to achieve its goals. For this project, they are centered on the core purpose of the AR application

- **Marker-based recognition of textbook diagrams-** The application is designed to be a companion to the standard government-issued ICT textbooks in Sri Lanka. The system's primary input will be specific 2D diagrams printed in the Grade 10 ICT textbook. These illustrations will serve as "markers" that the mobile device's camera recognizes to trigger the corresponding AR experience. This approach ensures a direct alignment with the national curriculum and classroom practice. The application must be able to accurately recognize these markers and overlay the 3D simulations with minimal latency. This is a similar approach to other AR apps developed for educational purposes.
- **Interactive 3D ICT simulations-** The software's core function is to transform static, two-dimensional diagrams into dynamic, interactive 3D models. For instance, a static diagram of a logic gate could become a working simulation, or a network topology diagram could visualize the flow of data packets. The system must render these principles as manipulable 3D objects, allowing students to use simple controls like sliders or gestures to change variables and see real-time changes in the simulation. This interactive and inquiry-based approach is intended to make abstract concepts tangible and improve comprehension.
- **Offline functionality-** To ensure the application is practical and accessible in Sri Lankan schools, many of which have unreliable internet connectivity, it is designed to be fully functional without an internet connection. All necessary assets, including 3D models, textures, and scripts, will be preloaded onto the device during the initial installation.
- **Teacher support guides-** To bridge the "pedagogical integration gap," the application will include teacher-centered support materials. This includes quick-access guides within the application interface with tips on how to use the AR tool in lesson plans. These materials are crucial for helping teachers who may be unfamiliar with digital pedagogy to integrate the technology effectively into their instruction.

4.2 Non- Functional Requirements

Non-functional requirements define the quality attributes of the system. For this project, they are heavily influenced by the specific constraints of the target environment.

- **Smooth performance on low-end Android devices-** The application is being developed with the specific hardware limitations of the Sri Lankan context in mind, where many students have access to low-end Android smartphones with limited memory (e.g., 2–3 GB RAM). Therefore, the system must be optimized to operate smoothly on these devices, with a target marker recognition response time of two seconds or less in typical classroom lighting conditions.
- **Usability- Simple UI for 15–16-year-olds-** The user interface must be intuitive and user-friendly, with minimal text and simple controls that are appropriate for a Grade 10 student. The design will adhere to UI/UX best practices, using clear instructions and easily identifiable buttons and icons to ensure smooth and engaging user experience.
- **Reliability across classroom environments-** The application must perform reliably in a variety of real-world classroom settings. This includes adapting to varying lighting conditions and situations where the textbook diagrams may be partially obscured. The system should also include a mechanism to provide helpful feedback, such as "Marker not detected, adjust camera," to guide the user if recognition fails.
- **Security-Minimal data collection, offline-first design-** The application is designed to be privacy-conscious. It will not collect or transmit any private student information without explicit consent. The offline first design inherently supports this by minimizing data exchange, making it a secure solution for environments with low data connectivity and strict privacy rules.

5 DESCRIPTION OF PERSONAL AND FACILITIES

- Academic Background and Educational Foundation

As a final-year undergraduate student at the Sri Lanka Institute of Information Technology (SLIIT), I am pursuing a Bachelor of Science (Hons) degree in Information Technology with a specialization in Interactive Media. This specific program has provided me with a comprehensive foundation that directly aligns with the technical requirements of this augmented reality educational project. My coursework has encompassed critical areas including 3D modeling and animation, where I learned to create and optimize three-dimensional objects for real-time applications. Game development modules have taught me the principles of interactive design, user experience optimization, and performance considerations essential for mobile applications. Additionally, my studies in user interface and user experience (UI/UX) design have equipped me with the skills necessary to create intuitive interfaces that can be easily navigated by Grade 10 students, ensuring the application remains accessible and engaging rather than technically overwhelming.

The Interactive Media specialization has been particularly valuable in preparing me for this project because it bridges the gap between technical implementation and creative content creation. Through various assignments and projects, I have learned how to balance aesthetic appeal with functional performance, a crucial skill when developing educational applications that must maintain student engagement while running efficiently on resource-constrained mobile devices. My academic training has also emphasized the importance of user-centered design, teaching me to consider the end-user experience from the initial concept through final implementation, which is essential when creating educational technology that must serve both students and teachers effectively.

- Technical Competencies and Hands-on Experience

My technical skill set has been developed through both formal coursework and practical application in personal and academic projects. I am proficient in Unity 3D development, having used this platform extensively for creating interactive applications and games during my studies. My experience with C# programming language, which is Unity's primary scripting language, has been developed through multiple projects where I implemented game mechanics, user interfaces, and data management systems. This programming experience is directly applicable to the augmented reality application, as it involves creating interactive behaviors for 3D objects, managing camera inputs, processing image recognition data, and optimizing performance for mobile devices.

My experience with 3D modeling tools, particularly Blender and Maya, enables me to create the custom 3D models required for visualizing ICT concepts. This includes designing simplified yet accurate representations of computer hardware components, network topologies,

and abstract processes like data flow that must be rendered efficiently on mobile devices. Understanding the technical constraints of mobile 3D rendering, such as polygon count limitations and texture optimization, is crucial for ensuring the application performs smoothly on the 2-3GB RAM Android devices commonly used by Sri Lankan students. My familiarity with augmented reality development principles, gained through academic projects and personal exploration, provides me with the foundational knowledge needed to implement marker-based recognition, camera calibration, and real-time 3D overlay systems.

Additionally, I have developed practical skills in mobile app development considerations, including understanding Android development requirements, app store deployment processes, and mobile-specific optimization techniques. My experience with version control systems like Git ensures that the development process will be well-managed and documented, which is important for a research project that must demonstrate systematic development and testing procedures.

- External Support and Professional Consultation

Recognizing that technical proficiency alone is insufficient for creating effective educational technology, I have established connections with local educational professionals who will provide crucial guidance throughout the project development process. I will be consulting regularly with experienced ICT teachers from government schools who have direct classroom experience with the Grade 10 curriculum I am targeting. These educators understand the specific challenges students face when learning abstract ICT concepts and can provide insights into which topics are most difficult to grasp using traditional textbook methods.

These teaching professionals will serve as subject matter experts who can review the accuracy of my 3D visualizations and simulations, ensuring that the augmented reality content correctly represents the concepts being taught. They will also provide feedback on the pedagogical effectiveness of the interactive elements I design, helping me determine whether the AR experiences actually enhance understanding or simply provide entertainment without educational value. Their classroom experience is invaluable for understanding the practical constraints of implementing new technology in Sri Lankan schools, including typical lesson duration, classroom management considerations, and the varying levels of technical literacy among both students and teachers.

Furthermore, I plan to collaborate with educational technology researchers and curriculum development specialists who can provide guidance on evaluation methodologies and help ensure that my research design meets academic standards for measuring learning effectiveness. This external support network ensures that the project maintains both technical quality and educational relevance, addressing the critical gap between proof-of-concept technology demonstrations and practical classroom applications that can improve student learning outcomes in real-world educational settings.

6 PROJECT TIMELINE

Phase	Duration	Key Activities	Deliverables
Requirements & Research	Months 1-2	Conduct literature review, consult with ICT teachers and educational professionals, analyze Grade 10 syllabus for AR-suitable topics.	Requirements document, Research foundation, Finalized list of AR-enabled topics.
System Design	Months 3-4	Create wireframes and storyboards for user interactions, design the system architecture, create a visual style guide for 3D assets and UI.	System design documents, User flow diagrams, Initial wireframes.
Asset Creation & Modeling	Months 5-6	Model 3D assets for network hardware and other components, texture development; optimize models for low-end Android devices.	Complete 3D asset library, Optimized textures and models
Prototype Development	Months 7-8	Set up the Unity environment, implement marker recognition and basic AR functionality, integrate a select number of core simulations.	Working AR prototype, Basic functionality implemented.
Full Deployment & Data Collection	Months 9-10	Conduct quasi-experimental study with experimental and control groups; deploy the final application, collect quantitative data from pre- and post-tests and qualitative feedback from interviews.	Complete empirical dataset, Interview transcripts, System usage logs.
Data Analysis & Reporting	Months 11-12	Analyze collected data, compare learning gains between groups, synthesize findings into a final report, prepare materials for presentation and documentation.	Comprehensive final report, Presentation slides, Research outputs.

Table 1: Project Timeline

7 REFERENCES

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